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# Chapter 1 Introduction to Industrial Artificial Intelligence

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## 1.1 Fundamental Problems in Industry

Digital transformation is a journey to develop data-centric operations with great visibility and productivity. Through the fourth industrial revolution efforts, companies can generate substantial financial and operational advantages, improving productivity and increasing customer satisfaction. The fundamental problem is digital transformation is to address productivity issues on flexibility, quality, and speed. Flexibility is achieved through the collaboration of machines and humans, forming a responsive, on-demand production system capable of adapting in real time. Quality is enhanced through real-time plant monitoring and the application of just-in-time maintenance. The degradation of manufacturing equipment and tools affects product quality and reduces productivity by increasing the occurrence of unplanned downtime. Thus, intelligent prognostic and health management (PHM) tools are essential for timely maintenance, guaranteeing the provision of high-quality products, minimizing unplanned downtime, and enhancing customer satisfaction. Speed is attained by improving interconnectivity between different sectors within the manufacturing process, impacting the entire product lifecycle. The synchronization and integration of data across companies, both vertically

and horizontally, promotes transparency and cohesion across departments and functions, greatly boosting manufacturing efficiency.

However, many production sites have found it difficult to quickly implement these technologies on a broad scale due to three major problems including:

- Discipline Problems: These problems include workforce competency, organizational culture, and managerial competencies. Japan is a perfect example of a trained workforce and effective mechanisms to transfer knowledge that can foster a strong company culture.
- System Problems: These problems refer to equipment, systems, and processes. Germany exhibits superior skill by employing precision-engineered equipment, rigorous process standards, and advanced design and manufacturing capabilities to facilitate knowledge transfer.
- Intrinsic Problems: These problems entail creating value for customers. The United States takes the lead in pioneering innovative business models and deploying technologies, employing collaborative innovation that leverages intellectual property, domain data, and ongoing service innovations for knowledge transfer.

Classical machine learning (ML) has created broad use cases to address these problems across manufacturing, such as predicting and preventing defects and failures. But generative AI is just beginning to be explored. Industrial AI emerges as a potent solution to these difficulties, providing substantial benefits for manufacturing improvements by enhancing the quality, structure, and vital aspects of manufacturing processes. The ability to train machines on large and unstructured datasets unlocks an entire suite of knowledge bases that can ultimately mimic human problem-solving capabilities using domain-based and data-rich sources in manufacturing processes. By standardizing workflows with data, industrial AI promotes the rapid accumulation of experience and aids in effective knowledge transfer. The strategic use of data not only uncovers latent problems in manufacturing systems but also supports transparent management of equipment health, stabilizes process parameters, and boosts overall efficiency. Furthermore, data serve as a medium for increasing user value, improving the functionality and reliability of products and equipment, enhancing operational efficiency, and bolstering the sustainable profitability of enterprises. Thus, incorporating industrial AI into manufacturing tackles these three critical problems through the application of data-driven insights and automation. It addresses discipline problems by boosting workforce competency and management skills with intelligent insights. System problems are alleviated by enhancing equipment design, manufacturing processes, and system integration, thereby making the en-

tire manufacturing process more transparent and effective. Additionally, intrinsic problems associated with customer value creation are solved by evolving business models and technologies, which foster collaborative innovation and continuous service enhancement. In summary, the necessity for AI intervention in the industry is highlighted by these challenges, demonstrating industrial AI's vital role in driving manufacturing excellence.

## 1.2 The Purpose of Industrial AI

The interpretation of industrial AI systems varies between academic and industrial sectors. Definitions of industrial AI as a unique technology or solution sometimes overlook critical questions such as the precise settings of intelligence required in industrial environments, the unsolved problems and challenges that current methods do not resolve, and the function of AI in overcoming these deficiencies. Furthermore, the role of industrial AI should not be limited to merely displaying the capabilities of data scientists in revolutionizing conventional industrial models. Instead, it should concentrate on uncovering and addressing the hidden problems within industrial ecosystems.

Industrial AI goes beyond merely adapting general AI technologies to industrial environment settings. The distinct characteristics of these settings – such as their fragmented nature, individualized challenges, and need for specialization – demand an integrated approach that merges computer science, AI, and domain-specific knowledge. Unlike conventional rule-based or mechanistic approaches, the true strength of data-driven industrial intelligence lies in its predictive analytics capabilities. These capabilities are founded on insights and evidence derived from data that facilitate the creation of intelligent management tools for tackling previously unknown challenges. Additionally, it assists in revealing complex interdependencies, thereby fostering the generation of new knowledge and supporting the evolution of a system that intelligently adapts and improves over time.

Challenges of industrial systems (Lee et al. 2019) can be generally divided into two main areas as depicted in Figure 1.1. Visible problems include issues such as machine failures, reduced production yield, and deteriorating product quality. On the other hand, invisible challenges encompass aspects like machine wear, component deterioration, and inadequate lubrication. Commonly identifiable problems such as equipment malfunctions, quality defects, and productivity declines are typically managed through continuous improvements and standardized practices, reflecting

the conventional approach to manufacturing (located in the lower left quadrant). Modern manufacturers are increasingly adopting AI algorithms to gain a competitive advantage. This approach is directly toward designing, producing, and delivering high-quality products more rapidly than competitors, thereby focusing on problem-solving (located in the upper left quadrant). Recent efforts by various companies have led to the development of new methods and techniques aimed specifically at addressing invisible challenges (positioned in the lower right quadrant). The adoption of an industrial AI-driven approach offers the potential to open up new opportunities for value creation in smart manufacturing, especially in environments that are dynamic and unpredictable (found in the upper right quadrant). Extensive implementation of industrial AI's fundamental components not only aids in addressing visible problems but also in avoiding invisible ones.

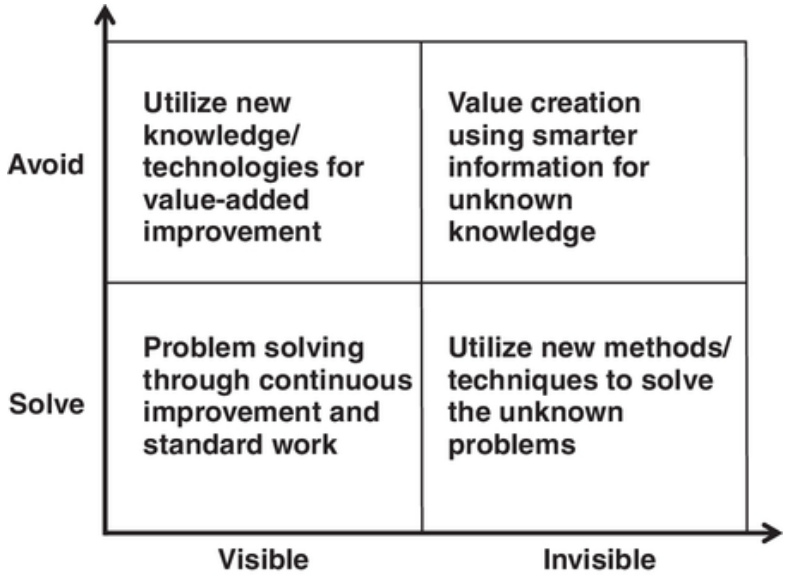



Figure 1.1 Visible and invisible problems in industrial systems and opportunities for industrial AI.

Industrial AI also plays a critical role in achieving the 3W's of smart manufacturing, namely work reduction, waste reduction, and worry-free manufacturing. The idea of "worry" in modern manufacturing systems is frequently derived from invisible difficulties such as machine degradation, process variation, and operation uncertainties. To address these difficulties, it is critical to implement industrial AI technologies in a systematic approach. Furthermore, the aim of reducing workloads and waste may be achieved by identifying the visible aspects of these difficulties and proactively resolving their possible future consequences using adaptive AI modules.

## 1.3 Difference Between AI and Industrial AI

There are clear differences (Lee 2020) between AI in general and industrial AI. These differences extend beyond the domain of application and include differences in functional requirements and algorithmic techniques. Before diving into these differences, it is necessary to define both terms. AI is a field of cognitive science that includes considerable study in image analysis and machine vision, natural language processing (NLP), robotics, and ML, among other topics. Despite its intense perspective, AI is usually veiled in mystery, with critics citing a lack of verifiable proof to back up its usefulness, repeatability, and financial return on investment in an industrial environment.

Conversely, industrial AI is described as a systematic discipline for the creation, validation, and rapid deployment of ML algorithms customized for industrial use cases, resulting in sustained performance. This domain focuses on developing intelligent systems for industrial applications that incorporate systematic development, quick deployment, and sustainability. Because of its emphasis on convergence and efficiency, industrial AI plays an important role in improving energy efficiency, industrial production and equipment safety, transportation safety, and operation resilience. Its uses are mostly focused on industrial systems including aerospace, transportation, energy, semiconductor, electronics, and automation processes (see Table 1.1).

TABLE 1.1

Comparison of AI and industrial AI.

| | AI | Industrial AI |
|---|---|---|
| Definition | A trial-and-error judgment-driven technology applied to NLP, image processing, automatic reasoning, robotics, and so on. It can be used in lots of areas like medicine and business. However, it hasn't worked out well in engineering yet. | In the industrial domain, the deployment of AI applications is aided by training and methodology that are distinguished by their speed, methodical approach, and sustainability. This framework assures that a varied variety of users may efficiently utilize the tools to achieve consistent outputs, thereby contributing to the evolution of AI standardization. |
| Function | Divergent and opportunity-driven situations (e.g. autonomous driving, economy sharing, and facial recognition) | In circumstances when we want to grow to be better and more efficient, we start with what we have and strive to improve it in a variety of ways. This includes increasing the speed and quality of manufacturing, using less energy to save money, making machinery more dependable, and making automobiles safer. |
| Application Areas | Social networks, financial business, medical field, among others | Multiple industrial applications include industrial equipment and manufacturing, power grids, power generation equipment, transportation and logistics, healthcare systems, and so on. |

| | AI | Industrial AI |
|---|---|---|
| Algorithms | ML, deep learning (DL) | Nontraditional ML (i.e. topology, transfer, learning, domain adaptation, and similarity-based learning), DL, broad learning, fuzzy learning, and augmented learning. |

To develop industrial AI with scalability, the following 5S features must be met:

- Systematic Approach: The creation of precise and standardized protocols to categorize the task boundaries and interfaces of Industrial Intelligence at various levels is necessary for the systemization of technology and application levels. A methodical strategy is needed to use the entire system to accomplish value enhancement, from the component, equipment, system, and community levels.
- Standardization Process: Standardization includes not just protocols, but also methodology, process, measurement, modeling processes, data quality, model assessment, fault-tolerant mechanisms, prediction-based operating procedures, and uncertainty management, among other things. It is particularly vital to standardize the decision-making process. If something cannot be integrated with existing standards, it will be difficult to incorporate technology into the industry and create value.
- Streamlining Process: The development and implementation of Industrial Intelligence systems follow a structured approach, involving the process whereby the intelligent industrial system receives inputs and produces decision-making outputs, as well as the interconnection of the industrial system to facilitate the swift deployment of intelligent applications.
- Speed of Realization: Although challenges in industrial systems are evident, numerous systems exist that can swiftly construct, model, validate, and deploy solutions to address fragmentation issues and fulfill the customized needs for rapid response.
- Sustainable Capability: Just as there is a need for clarity and reliability in AI predictions, sustainability means consistently obtaining the same model outcomes from a given data set. Lacking this clarity, it becomes challenging to maintain standardization and uniformity in manufacturing systems.

The aim of standardization, systemization, and streamlining is to attain both speed and sustainability. Each has its unique manufacturing ideologies to accomplish this. In Japan, insights from problem-solving are transmitted through training to solve and avoid issues. In Germany, solutions to problems are integrated into equipment, which then serves as a vessel for knowledge to automatically solve and prevent issues. In the United States, problems are comprehended through data analysis, leading to knowledge acquisition, problem resolution, and redefinition, thereby creating a closed-loop system. The subsequent discussion delineates the distinctions among industrial AI, ML, expert systems, and expert experience. Expert experience derives from the long-term practice of technicians, as exemplified by fault diagnosis in rotating machinery, where seasoned experts can precisely identify issues using just their hearing. However, this type of experience poses significant challenges in knowledge transfer, often disappearing with personnel changes, thereby affecting the sustainability of enterprise problem-solving. Moreover, it has limitations with invisible problems, such as assessing the current health of equipment, which even experienced experts cannot determine accurately. Expert systems, which rely on a knowledge base and inference engines, struggle with uncertainties due to environmental and operational changes. These systems require regular updates to enhance performance. Contrastingly, systems developed through AI and ML offer markedly improved problem-solving accuracy and possess impressive learning capabilities compared to the aforementioned methods. However, they lack robustness under variable working conditions and diverse data sets. Future industrial AI, leveraging multidimensional thinking and nontraditional ML techniques such as transfer learning, domain adaptation, stream-of-x ML similarity-based ML, etc., to manage the system performance for highly connected and highly complex industrial systems.

## 1.4 Definition and Meaning of Industrial AI

Industrial AI (Lee et al. 2018, 2020) marks a pivotal transformation in how various industries operate, ushering in a new era characterized by enhanced efficiency, reliability, and productivity. This shift is largely fueled by the increasing demands for predictive maintenance, the streamlining of production processes, and the optimization of manufacturing systems. The genesis of industrial AI was spurred by the recognition of the need to systematically integrate AI technologies into industrial settings. This specialized approach sets it apart from general AI applications that encompass everything from autonomous vehicles to virtual assistants. Tailored specifically for industrial use, industrial AI focuses on boosting operational efficiency, reliability, and productivity, effectively

tackling the unique challenges and complexities found in these environments.

At its heart, industrial AI involves using ML algorithms and AI techniques to improve industrial processes. This includes a thorough process of developing, validating, and implementing these technologies to solve practical industrial problems, bridging the gap between academic research and industrial use. Industrial AI goes beyond simply adapting general AI technologies; it requires a deep integration of AI with specific industry knowledge to address the unique challenges of industrial environments. The main goal of industrial AI is to enhance the performance and reliability of industrial systems through predictive analytics. This leads to lower operational costs and increased overall efficiency. By utilizing various data sources such as sensor data and historical records, industrial AI can predict equipment failures, optimize maintenance schedules, and boost plant performance. This systematic approach ensures the long-term sustainability and continual value of industrial AI solutions.

Industrial AI relies heavily on extensive data sets to train its algorithms. Companies that have embraced industrial digitalization already have a constant stream of data flowing from their assets. However, much of this data, while useful, may not be suitable for ML due to issues with its completeness, lack of baseline information, and inconsistencies. According to a recent report by the National Academy of Engineering, the main challenges in developing and deploying AI for industrial use include the poor quality of data and the absence of a systematic method for developing and validating AI technologies. Moreover, concerns about AI trustworthiness and the absence of standardization present additional risks for AI applications in the industry.

The primary challenge involves establishing a systematic discipline that focuses on the development, validation, and deployment of AI methods and ML algorithms for industrial applications, ensuring their sustainability, scalability, and security. This involves integrating advanced sensing, communication, and big data analytics platforms, enabling the fusion of physical systems with computational models through an integrated learning platform. This systematic approach in industrial AI is aimed at addressing three key areas – data, domain, and discipline – each critical to transforming traditional AI methodologies into robust interdisciplinary capabilities across varied engineering systems:

- Data: Emphasizes foundational research underpinned by systematic thinking, involving learning in ML, data engineering, and advanced

analytics. This foundational approach ensures that data is not only collected but also effectively analyzed and utilized to drive innovation.
- Domain: Focuses on industry-inspired research, learning, and application in diverse fields such as smart and digital manufacturing, energy systems (e.g. electrification and wind turbines), healthcare and medical sectors (including ICU, TBI, and Pharmacy 4.0), semiconductor manufacturing, and the development of industrial AI system engineering platforms. This area encourages the application of AI technologies tailored to specific industry needs and challenges.
- Discipline: Dedicated to methodological innovation and education, forming the core concept of industrial AI. This involves providing comprehensive methodologies, tools, and curricula designed to foster innovation and practical application in various disciplines.

These three areas form the backbone of a systematic industrial AI methodology, facilitating a significant transformation from common practices in traditional AI and ML toward developing capabilities that strengthen interdisciplinary engagements in diversified engineering systems.

## 1.5 Key Elements in Industrial AI: ABCDE

The core of industrial AI is established on five key elements, collectively known as ABCDE. These components are vital in the creation and utilization of industrial AI technologies, which facilitate the transformation of industrial processes into systems that are smart, efficient, and predictive.

- Analytics: This is central to industrial AI, comprising the methods and technologies applied to analyze data and derive valuable insights. This component encompasses the deployment of ML algorithms, statistical models, and data processing techniques to uncover trends, behaviors, and results in industrial operations. Through analytics, industries can progress from basic descriptive analysis to more advanced predictive and prescriptive insights, facilitating proactive decision-making and enhancing process optimization.
- Big Data Technology: This supplies the extensive and diverse datasets critical for industrial AI functionality. Originating from sensors, machinery, and operational workflows in high volumes, this data is distinguished by its vast volume, rapid velocity, and wide variety. It is crucial for training AI models, detecting patterns, and enabling data-driven decisions. Big data technologies play a key role in the storage, processing, and analysis of industrial data, ensuring the efficient performance of industrial AI systems and providing real-time insights.

- Cloud or Cyber Technology: This provides the essential infrastructure and platform for deploying and accessing industrial AI applications. It enables the scalability, flexibility, and computational power required to handle big data effectively. Through a centralized platform for data storage, analytics, and application deployment, cloud technology facilitates the integration of AI into industrial operations and supports the creation of CPS. These systems enable the monitoring, control, and optimization of physical industrial processes using cyber technologies.
- Domain Knowledge: It is crucial for the effective application of industrial AI. It involves a thorough comprehension of the unique needs, challenges, and complexities specific to the industrial sector where AI is applied. This understanding ensures that AI solutions are practical, relevant, and effective in addressing real-world industrial problems. Domain experts are integral to the development of industrial AI applications, ensuring that they conform to the practicalities of industrial operations and satisfy the specific demands of the industry.
- Evidence: This includes the data and empirical results that validate the accuracy and effectiveness of industrial AI models. Gathering and analyzing performance data is essential to confirm that AI applications achieve their objectives, such as increasing efficiency, reducing downtime, or enhancing quality. Evidence is crucial for building confidence in industrial AI systems, proving their worth, and encouraging their broader adoption across industries. Moreover, it plays a key role in the ongoing improvement and fine-tuning of AI models, ensuring their sustained relevance and effectiveness.

The ABCDE framework presents a comprehensive approach to the development and implementation of industrial AI systems. By focusing on these essential elements, industries can harness the power of AI to transform their operations, realize substantial efficiencies, and gain a competitive edge in the digital and automated industrial environment. The effective integration of these components requires a concerted effort from data scientists, domain experts, and industrial practitioners, to make sure that industrial AI solutions are technologically advanced, practically feasible, and closely aligned with industrial realities.

## 1.6 CPS Framework for Industrial AI

CPS represents the fusion of physical industrial processes with sophisticated cyber technologies. This integration facilitates the monitoring, controlling, and optimizing of industrial operations via real-time data analytics and AI-powered insights. CPS provides the essential framework for implementing industrial AI, enabling industries to utilize computational in-

telligence to improve efficiency, productivity, and flexibility in a rapidly digitizing world. The integration of CPS with a 5C architecture is crucial for advancing industrial AI (Lee et al. 2014).

The 5C architecture presents a structured approach for deploying CPS in industrial settings, enabling the phased incorporation of industrial AI throughout different operational layers. This architecture includes five critical components, each targeting distinct facets of CPS integration:

- Connection: Sets up the necessary data infrastructure to facilitate the seamless gathering of data from sensors and devices throughout the industrial network.
- Conversion: Converts raw data into useful features through preprocessing and sophisticated analytics, paving the way for informed ML.
- Cyber: Creates a digital twin of the physical system, providing a virtual environment for simulation and optimization that does not interfere with actual operations.
- Cognition: Utilizes AI and ML algorithms to visualize data, generating a strategic understanding of complex relationships for decision-making.
- Configuration: Implements the insights obtained to optimize, recalibrate, and reconfigure the physical processes, enhancing performance and efficiency in real time.

Several critical technologies reinforce the implementation of CPS and the 5C architecture. The Internet of Things (IoT) (Xu et al. 2014) connects physical devices and machinery, facilitating efficient data gathering and communication. Big Data Analytics (Lee et al. 2013, 2015; Shi et al. 2011) processes and analyzes vast amounts of data, producing insights that guide decision-making. Cloud Computing (Wu et al. 2013; Yang et al. 2015; Zhang et al. 2014) offers essential computational resources, data storage, and AI services, bolstering the cyber aspect of CPS. Specifically, ML and AI power the cognition component, enabling predictive analytics and smart automation.

CPS significantly enhances the implementation of industrial AI by providing a unified framework that integrates digital technologies with physical industrial processes. Through this framework and the 5C architecture, industries can achieve real-time monitoring and control, minimizing downtime and increasing operational efficiency. Predictive maintenance, powered by AI, proactively identifies maintenance requirements, reducing costs and boosting productivity. Strategic decision-making, supported by AI and analytics, enables precise, data-driven strategies that can adapt to changing operational conditions. Furthermore, by utilizing CPS and in-

dustrial AI, companies can accelerate innovation, enhance product quality, and strengthen their competitive position in the market.

The integration of the CPS framework, the 5C architecture, and essential enabling technologies is crucial for advancing industrial AI in the manufacturing sector. This comprehensive approach not only enhances operational efficiency but also drives innovation, securing a competitive edge for industries amidst fast-paced technological advancements. By adopting CPS and the principles of the 5C architecture, industries can navigate the challenges of digital transformation and capitalize on the opportunities offered by industrial AI.

## 1.7 Technological Elements of CPS Framework

Key enabling technologies essential for success in connection, conversion, cyber, cognition, and configuration are summarized in four categories: data technology, analytical technology, operation technology, and platform technology, detailed as follows:

- Data Technology: During each step of the manufacturing process, unique data is produced at the component, machine, or shop-floor level, which can be structured or unstructured. The “3B” data issues – broken data, bad baseline, and background missing – significantly affect the implementation of the CPS 5C architecture in industrial settings. It’s vital to achieve a unified collection of diverse data, automate the extraction of valuable insights, and standardize data acquisition. Additionally, data management and synchronization are key considerations. Core elements of data technology include sensor technology and data acquisition, data indexing, transmission, and storage, data quality evaluation, big data representation and visualization, and data security and privacy protection.
- Analytics Technology: Sensors and data acquisition systems are crucial in data technology for providing the extensive data required for analysis and valuable information extraction. Analytics technologies serve as a bridge between the physical and cyber worlds, aligning with the conversion layer of the CPS framework, and include both edge and cloud computing analyses. Although edge computing is well-suited for analyzing real-time data close to industrial sites, its limited storage and processing capacities make centralized model training and prediction challenging. Consequently, there is a need to focus on developing flow-based reasoning technologies that use real-time data streams to continually refine data relevance for model self-optimization. Additionally, exploring new modeling and analysis tools such as time

machine-based state modeling, semi-supervised, peer-to-peer, adaptive, and transfer learning is essential. These emerging techniques include signal processing, statistical analysis, detection, classification, prediction, clustering, relationship mining, causal analysis, and modeling of complex, dynamic systems.

- Operation Technology: Operation technologies, essential for decision-making and action-taking based on data, form a critical part of the cognition layer in the CPS 5C architecture. Used alongside three other technologies, they help optimize production processes. This involves three key aspects of operational management development: (i) effectively applying predictive model insights to maintenance operations, (ii) improving enterprise-level management decisions, and (iii) transitioning from experience-based to data-driven production methods.
- Platform Technology: Platform technologies serve as the foundation for all other enabling technologies, supporting the connection, conversion, and cyber layers in the CPS architecture. For edge computing platforms, it is essential to enhance data acquisition and processing capabilities through hardware innovation and to develop edge platforms' collaborative capabilities for self-organization and self-configuration in production systems. Cybersecurity becomes vital due to the interconnected nature of these platforms, necessitating robust hardware architecture and software mechanisms to secure system access, particularly post-feedback control implementation on edge devices. For cloud platforms, key developmental areas include life cycle management, uncertainty management, and continuous self-learning models. Overall, platform technologies can be categorized into three service levels: infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS).

## 1.8 Developing Industrial AI Talents

To cultivate a new breed of industrial AI engineers, it's crucial to rapidly advance AI talent development. The industrial AI necessitates a data foundry to serve as a data hub, hosting various datasets from real-world industrial challenges, which helps train engineers to systematically apply AI in solving problems. These datasets encompass a wide array of manufacturing and engineering systems, such as automotive, aerospace, semiconductors, industrial automation, and so on.

The training of industrial AI talents can be structured around the 4Ps approach, which includes:

- Principle-based Learning: This involves understanding fundamental principles, concepts, and knowledge, along with developing the ability to learn independently.
- Practice-based Learning: Learners apply knowledge and tools to solve specific problems using the assigned datasets.
- Problem-solving Learning: This focuses on the planning of problem formulation with their own datasets to train the ability to design, develop, implement, and optimize systems.
- Professional-based Learning: This stage involves defining the ecosystem of data issues within complex systems, designing a system engineering, and leading teams to tackle system development and implementation with sustainable performance.

Take smart manufacturing as an example, students will be taught to exploit industrial AI to uncover hidden correlations in operational conditions, using these insights to enhance uptime, productivity, and efficiency. For predictive maintenance, industrial AI can identify early system changes, estimate remaining useful life, and streamline maintenance activities to prevent operational interruptions. These skills can be continuously developed in the workplace, encouraging students to evolve into industrial AI leaders.

## 1.9 Training Industrial AI Talents Using Open-source Datasets

This section explores the practical application and deployment of the industrial AI architecture framework across various industrial settings. There are numerous in-depth reviews of AI and ML strategies tailored for various industrial applications (Jia et al. 2018; Kumar et al. 2023; Nguyen et al. 2023; Ochella et al. 2022; Qiu et al. 2023; Rezaeianjouybari and Shang 2020; Serradilla et al. 2022; Zhang et al. 2019). From these reviews, many existing research papers are evaluated not only qualitatively but also quantitatively, to offer unique insights and identify emerging trends and innovations in AI and ML strategies for industrial AI across different scenarios.

Addressing the three critical dimensions – data, discipline, and domain – in industrial applications is crucial. There is an immediate need for accelerated research, development, prototyping, and deployment of AI and ML-enabled solutions. Such advancements are expected to generate significant operational, technological, and economic benefits across a broad spectrum of industries. As a case study, we utilized industrial open-source datasets to demonstrate the significance of progress in industrial data an-

alytics and industrial AI. Table 1.2 lists 20 open-source industrial datasets released by the PHM Society and the IEEE Reliability Society from 2008 to 2023 for data challenge competitions (Jia et al. 2018; Su and Lee 2023). This table illustrates the various challenges when dealing with problems from various industrial data across different industrial settings. Understanding the challenges of real industry and the relevant ML methods to solve these problems in the past, helps to foster the development and validation of innovative AI and ML approaches, effectively narrowing the divide between academic research and industrial practice.

TABLE 1.2

Industrial open-source datasets released by PHM Society and IEEE Reliability Society.

| Data Challenge Competition | Task Description | URL |
|---|---|---|
| 2023 IEEE Gearbox | 1. Fault diagnosis | http://www.ieeereliability.com/PHM2023/assets/data-challenge.pdf |
| 2023 PHM AP Spacecraft System | 1. Fault diagnosis<br>2. Unknown case identification | https://phmap.jp/program-data/ |
| 2022 PHM NA Rock Drill | 1. Multiclass fault classification | https://data.phmsociety.org/2022-phm-conference-data-challenge/ |
| 2022 PHM EU Printed Circuit Boards | 1. Fault diagnosis | https://github.com/PHME-Datachallenge/Data-Challenge-2022 |
| 2021 PHM NA Turbofan Engine | 1. RUL prediction | https://data.phmsociety.org/2021-phm-conference-data-challenge/ |
| 2021 PHM EU Manufacturing Production Line | 1. Fault detection<br>2. Fault diagnosis | https://github.com/PHME-Datachallenge/Data-Challenge-2021 |

| Data Challenge Competition | Task Description | URL |
|---|---|---|
| | 3. Root cause identification | |
| 2020 PHM EU Filtration System | 1. RUL prediction | http://phmeurope.org/2020/data-challenge-2020 |
| 2019 PHM NA Fatigue Crack | 1. Crack length estimation | https://data.phmsociety.org/2019datachallenge |
| 2018 PHM NA Ion Mill Etch System | 1. Fault diagnosis<br>2. RUL prediction | https://www.phmsociety.org/events/conference/phm/18 challenge |
| 2017 PHM NA Bogie | 1. Fault detection<br>2. Fault diagnosis | https://www.phmsociety.org/events/conference/phm/17 challenge |
| 2016 PHM NA CMP | 1. MRR prediction | https://www.phmsociety.org/events/conference/phm/16 challenge |

| Data Challenge Competition | Task Description | URL |
|---|---|---|
| 2015 PHM NA Power Plant | 1. Fault detection<br>2. Fault diagnosis | https://www.phmsociety.org/events/conference/phm/15 challenge |
| 2014 IEEE Fuel Cell | 1. Health assess-ment<br>2. RUL predic-tion | http://eng.fclab.fr/ieee-phm-2014-datachallenge/ |
| 2014 PHM NA Unknown | 1. Risk assess-ment<br>2. Fault detection | https://www.phmsociety.org/events/conference/phm/14 challenge |
| 2013 PHM NA Unknown | 1. Faut detection<br>2. Fault diagnosis | https://www.phmsociety.org/events/conference/phm/13 challenge |
| 2012 PHM NA Bearing | 1. RUL predic-tion | https://www.femto-st.fr/en/Research-departments/AS2M/Research-groups/DATA-PHM |

| Data Challenge Competition | Task Description | URL |
|---|---|---|
| 2011 PHM NA Anemometer | 1. Fault detection | https://phmsociety.org/phm_competition/2011-phm-soc conference-data-challenge/ |
| 2010 PHM NA Milling Cutter | 1. Health assess-ment | https://phmsociety.org/phm_competition/2010-phm-soc conference-data-challenge/ |
| 2009 PHM NA Gearbox | 1. Label fault samples<br>2. Fault diagnosis | https://phmsociety.org/public-data-sets/ |
| 2008 PHM NA Aircraft Engine | 1. Engine RUL predic-tion | https://c3.ndc.nasa.gov/dashlink/projects/15/ |

Table 1.2 presents that the industrial open-source datasets encompass a wide range of systems, including aircraft engines, milling cutters, bearings, anemometers, fuel cells, power plants, CMP machines, bogies, ion mill etching tools, filtration systems, manufacturing production lines, turbofan engines, printed circuit boards, rock drills, spacecraft propulsion systems, and gearboxes (Jia et al. 2018; Su and Lee 2023). The challenges that these datasets faced primarily focus on fault detection, diagnosis, health assessment, and prognosis (specifically predicting the remaining useful life), aiming to enhance industrial system operational efficiency, safety, and reliability. To facilitate discussion of various industrial open-source datasets, we have implemented a simplified naming convention: "YEAR ORGANIZATION" to denote the year and the organizing body of the competition for the specific industrial dataset. Among the competitions

reviewed, eighteen were hosted by the PHM Society, while the other two were managed by the IEEE Reliability Society. Meanwhile, “PHM NA,” “PHM EU,” and “PHM AP” are used as abbreviations to denote the competitions held in North America, Europe, and Asia-Pacific by the PHM Society, respectively. Additionally, “IEEE” is used to denote the competitions organized by the IEEE Reliability Society.

When analyzing AI and ML approaches to solve problems using those industrial open-source datasets, we divided them into two parts based on the tasks they faced - fault detection and diagnosis, health assessment, and prognosis. In the twenty data challenge competitions analyzed, twelve required participants to focus on fault detection and diagnosis. For fault detection and diagnosis problems, typically, this represents a classification problem, which is required to achieve anomaly identification, failure mode determination, and root cause analysis. It plays an important role in constructing a reliable and safe industrial system, ensuring effective maintenance interventions in a timely manner, and finally reducing downtime costs. Challenges commonly encountered in fault detection and diagnosis include accurately classifying multiple fault modes, handling missing data, addressing data imbalance, managing limited data, recognizing unknown faults, and compensating for domain shifts. A variety of ML solutions are employed to tackle these issues, ranging from statistical methods to conventional ML and DL techniques. For the classification problem, one of the important factors that significantly influence the choice of the modeling approach is the volume of data in the industrial open-source dataset. If the volume of the training data is large, researchers are in favor of DL approaches to develop data-driven models. For example, the large datasets available in the 2022 PHM NA and 2023 IEEE competitions have enabled the application of DL methods. Various convolutional neural network (CNN)-based methods (such as ensemble-based CNNs, inception-based CNNs, and residual-based CNNs), long short-term memory (LSTM) based methods (including Bi-LSTM, hybrid-LSTM, LSTM-FCN), and deep tree-based methods (like XGBoost and LightGBM) have been developed to tackle these challenges (Su and Lee, 2023). Conversely, conventional ML methods are more commonly utilized when the volume of the training data is small. For instance, in the 2021 PHM EU, 2022 PHM EU, and 2023 PHM AP competitions, where dataset sizes are relatively small, methods such as KNN, K-Means, similarity-based approaches, rule-based methods, Naïve Bayes, decision trees, and random forests are employed to effectively address these challenges (Jia et al. 2018; Su and Lee 2023). In the remaining eight competitions, the focus is on health assessment and prognosis. These tasks are crucial for evaluating industrial systems about their current health condition and predicting

their remaining useful life. Implementing proactive and predictive maintenance can help to check the system degradation in a timely manner and try to avoid unplanned downtime, ultimately achieving a longer lifespan of the industrial systems. The challenges commonly encountered in health assessment and prognosis include identifying degradation trends, estimating remaining useful life under various operating conditions, addressing data imbalance, and multiple fault modes.

Given the long-time series nature of the datasets involved in these tasks, various approaches to data utilization and augmentation are viable. The effective size of the training data can be significantly enlarged by reconsidering partial time series data points as a training sample. Then, even when industrial open-source datasets are really small (less than 100 samples), the size of the training data can also achieve over 10 000. This gives the possibility for the deployment of both conventional ML and DL methods. There has been a noticeable shift toward deploying more DL approaches rather than conventional ML approaches in these industrial open-source datasets over the years. Before 2015, statistical methods and traditional ML techniques were commonly used, including Particle Filters, Kalman Filters, similarity-based methods, self-organizing maps (SOM), Gaussian processes, support vector regression, and multilayer perceptron (MLP) (Jia et al. 2018). After 2015, more DL approaches like RNN- and CNN-based methods are surging in solving industrial application problems. This shift is attributable to the proficiency of RNN-based methods (such as LSTM, GRU, Bi-LSTM, and LSTM-MR) in handling sequential, long time-series data, and the capability of CNN-based methods to extract useful features, thereby enhancing the effectiveness of tackling assessment and prognosis challenges (Su and Lee 2023). Moreover, various transfer learning and domain adaptation techniques are also proposed to address the situation when the data distribution between the source domain and target domain is different (Su and Lee 2023).

Overall, the diversity of ML algorithms underscores their effectiveness in leveraging data-driven approaches to address challenges presented by these industrial open-source datasets. This versatility facilitates the broader adoption of ML and AI techniques in the industrial sector, highlighting the significant potential of industrial AI to transform industrial practices.

## 1.10 Issues in Industrial AI

While different industrial sectors have made significant strides in integrating AI into their operations, it continues to face substantial chal-

lenges. The evolution of sensor technologies, communication protocols, and computing power has dramatically enhanced data and platform technologies, paving the way for the development of advanced analytical tools. As these tools begin to demonstrate their potential, operational technologies are being refined and deployed. However, the widespread adoption of industrial AI in manufacturing is primarily hindered by two factors: the inherent characteristics of manufacturing operations, which rely heavily on large, costly, and often outdated hardware, and the hesitancy of senior management to embrace new technologies in established processes. Additionally, technical hurdles associated with enabling technologies demand attention. The expectations from industrial AI are vast and diverse, and even partial achievement of these expectations would highlight the significant and unique challenges of applying AI to industries. Among the existing challenges and complexities, the following are crucial issues that need to be addressed for the future of industrial AI.

- Cybersecurity: It is essential to protect data integrity, privacy, and confidentiality. Various threats can occur at different communication layers or during the transmission of data, with devices connected to the internet often serving as the initial points of entry for cyber attacks. These threats may involve compromised gateways, malware, and cross-site scripting attacks. At the smart nodes layer, integrity attacks are focused on corrupting sensor signals or the values controlling machines. Replay attacks, which involve the repeated transmission of valid data packets, are a significant risk here. The network layer is vulnerable to communication delays that can disrupt real-time services, with replay attacks again posing a notable threat. In the fog layer, attacks can lead to service unresponsiveness or denial, which affects the functionality of the system. Applications that require high levels of synchronization might experience de-synchronization due to these attacks. The cloud layer, which stores valuable data, is susceptible to threats such as denial of service, data breaches, malware, and unauthorized access. The complexity of these threats requires strong defensive measures, such as firewalls and encryption. Blockchain technology is viewed as a promising approach to enhance security and privacy through its decentralized nature. A proposed unified three-layer architecture seeks to address these challenges effectively.
- Data Integrity: In the domain of manufacturing, where processes are inherently nondeterministic, maintaining data integrity is crucial. It is essential to ensure the accuracy, correctness, completeness, and unbiased nature of data, from its initial generation and acquisition through to its transfer and storage. The complexity and vast scale of modern applications make physical verification of data by users both impracti-

cal and cumbersome. Consequently, rigorous data integrity checks are necessary throughout the data lifecycle to maintain reliability. Mechanisms must be in place to facilitate ongoing maintenance and validation of data, ensuring it remains free from errors and bias in large-scale environments where manual checks are unfeasible.

- Data Management: As industries increasingly integrate AI into their operations, the challenge of managing the large volume and variety of data becomes more pronounced. The necessity for robust storage solutions that can accommodate an ever-growing amount of data, information, and models is paramount. Traditional relational database management systems are often inadequate for the needs of manufacturing enterprise solutions due to their lack of flexibility and scalability. Alternatives such as NoSQL databases, including Key-Value stores, along with distributed file systems like Hadoop Distributed File System (HDFS) and Google File System (GFS), offer valuable options for exploring more scalable and flexible data management solutions. These technologies are crucial for effectively handling the complex data demands of AI-driven environments.
- System Availability: Ensuring the reliability and availability of both hardware and software is crucial in manufacturing, not only to maximize the operational efficiency of manufacturing assets but also to guarantee the functionality of the enabling technologies themselves. Hardware components involved in sensing, communication, and computation must be both reliable and accurate. Similarly, software services and servers must remain operational even during security breaches, periods of high traffic, or heavy load conditions, all while ensuring data safety and providing crucial diagnostic information in the event of failures. This dual focus on hardware and software reliability is essential for maintaining continuous operational continuity and system integrity.
- Analytics: Transforming data into actionable insights remains a fundamental challenge in the domain of AI, particularly as its applications broaden. In manufacturing, where each scenario might differ slightly from the next, customizing solutions and maintaining human oversight to fine-tune, test, and validate models for complex situations is essential. While technologies such as transfer learning have advanced, the suitability of algorithms is often restricted to specific scenarios, requiring particular data quantity and quality standards, and defined applications. Despite increases in computational resources, the performance of algorithms does not necessarily scale proportionally. Developments in computational hardware like MapReduce must be leveraged to improve algorithm efficiency. Furthermore, the need for real-time insights poses additional challenges in analyzing high-veloci-

ty and high-variety data streams, with underlying issues remaining critical and unresolved.

- Data Quality: AI algorithms, including those used in ML and DL, require massive, clean, and unbiased data sets to function effectively. The quality of the data directly impacts the accuracy of the models, with any learning from flawed, inaccurate, or inadequate data sets potentially leading to incorrect downstream results. Collecting such high-quality, representative data often poses challenges due to privacy concerns, accessibility issues, or the scarcity of certain events. Moreover, ensuring that the data is well-labeled and clean is a labor-intensive process that is susceptible to human error. These factors underscore the critical need for rigorous data preparation and management to support the effective deployment of AI technologies.
- Model Interpretability: The interpretability of ML and DL models, especially those involving deep neural networks, is often compromised by their "black box" nature. This term reflects the opacity of their decision-making processes, which can be challenging for humans to understand. Such lack of transparency becomes a critical issue in sectors where trust and clarity in decision-making are important, including healthcare and finance.
- Multistage Manufacturing Systems: Analyzing operational parameters and sensor data in complex multistage manufacturing systems continues to present considerable challenges. The stream-of-quality (SoQ) framework (Lee et al. 2022) has emerged as a novel approach that facilitates an integrated evaluation of quality indicators throughout the manufacturing process. A recent study (Lee et al. 2024) also proposes a methodological innovation by implementing topological analytics within a 5-level CPS architecture specifically designed for SoQ assessments in smart manufacturing environments. The methodology supports real-time monitoring and predictive analytics, enabling the discovery of intricate relationships between quality attributes and process variables across diverse manufacturing stages. Additionally, this research highlights the application of topological graph visualization techniques for dynamically identifying and integrating new representative data in quality assessments, enhancing decision-making processes in multistage manufacturing systems.
- Industrial Large Knowledge Model (ILKM): The ILKM framework (Lee and Su, 2023) represents a significant advancement in integrating large language models (LLMs) with specialized industrial knowledge to enhance Industry 4.0 and smart manufacturing. Recognizing the limitations of LLMs trained primarily on general knowledge, the ILKM framework seeks to tailor these technologies to meet the specific, complex demands of industrial applications. By incorporating advanced

AI, ML, and domain-specific insights, the ILKM aims to significantly boost the intelligence, efficiency, and resilience of future industrial systems.

## 1.11 Conclusion

The exploration of industrial AI has elucidated its transformative impact on future industries. From the inception of basic industrial challenges to the specialized applications in predictive maintenance and operational optimization, industrial AI emerges as a robust solution to enhance efficiency and reliability. The introduction of the ABCDE framework underscores the foundational elements – *Analytics, Big Data, Cloud technology, Domain knowledge, and Evidence* – that are essential for the practical application of industrial AI. The paper highlights the critical role of system integration, as demonstrated by the development and implementation of CPS within the manufacturing sector, which facilitates real-time monitoring and data-driven decision-making. This chapter also highlights the key technologies enabling the CPS framework, emphasizing the importance of data, analytical, operational, and platform technologies. These technologies support the robust implementation of CPS, ensuring that the connection, conversion, cyber, cognition, and configuration layers of industrial AI are effectively managed to optimize production and maintenance operations. Additionally, the challenges presented in implementing these technologies, such as data integrity and cybersecurity, emphasize the need for stringent standards and innovative solutions to safeguard and maximize the utility of industrial data.

The necessity for tailored AI training programs that develop industry-ready talent underscores the practical and educational shifts required to meet the demands of Industry 4.0. Moreover, the discussions around the potential and limitations of current AI technologies suggest that continuous advancements in AI methodologies are critical to overcoming the challenges of data quality and model interpretability. This comprehensive analysis culminates in the acknowledgment that while industrial AI holds significant promise for revolutionizing industrial operations, its success is dependent on multidisciplinary efforts that extend beyond technology to include rigorous data management, continuous innovation, and strategic partnerships, paving the way for future advancements in the field of smart manufacturing.

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